

Order Coleoptera

Beetles

About one in 3 insects in existence today is a beetle. Collectively they make up an order containing 370,000 known species in 166 families, and have successfully colonized every sort of habitat on land and in fresh water. They range in size from insects that are just visible to the naked eye to tropical giants 7 in (18 cm) long. Beetles are distinguished by their toughened front wings, or elytra, which fold over their membranous hindwings like a case. Because the hindwings are protected, beetles can squeeze into confined spaces. Most species are herbivorous, but the order also includes many predators and scavengers, and some parasites. Beetles develop by complete metamorphosis: there are distinct larva and pupal stages.

Family Anobiidae

Woodworm

Occurrence 1,500 spp. worldwide, especially in temperate regions; in wood, warehouses, stores, houses



These small, elongated to oval beetles are pale or reddish brown to black, and are mostly 1/16–1/4 in (2–6 mm) long. The head is often hooded by the pronotum. Eggs are laid in suitable food, into which the larvae bore tunnels. Several species are pests of cereals, legumes, spices, and tobacco.

PTILINUS PECTINICORNIS

Found in the UK and central Europe, this species is especially associated with beech trees and may attack wood furniture.



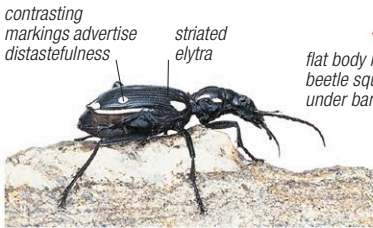
Family Carabidae

Ground beetles

Occurrence 29,000 spp. worldwide; on ground, under stones, logs, in debris, leaf litter, tree foliage

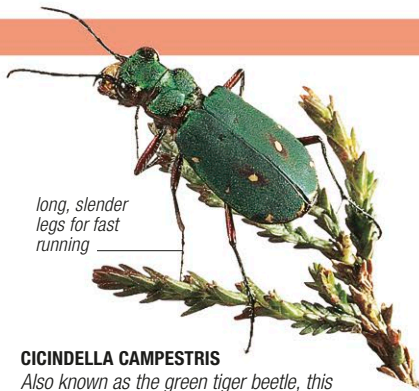


Measuring 1/16–3/4 in (0.2–8 cm) long, these elongated and flattened beetles can be dull or shiny, brown or black, often with a metallic sheen. The head, thorax, and abdomen are clearly differentiated, and the legs are adapted for quick escape from predators. A few species can deter predators using blasts of hot, caustic chemical substances expelled with an audible “pop” from the end of their abdomen. Most species are nocturnal hunters; adults and larvae are mostly predatory. Eggs are laid on the ground, on vegetation, and on decaying wood and fungi.



THERMOPHILUM SEXMACULATUM

Commonly called the domino beetle, this Afrotropical species hunts during the day in semiarid areas.



CICINDELLA CAMPESTRIS

Also known as the green tiger beetle, this fast-running, diurnal species can be found from April to September on sandy soils in Europe.



MORMOLYCE PHYLLOIDES

The violin beetle is found in the tropical forests of Southeast Asia, and feeds on insect larvae and snails.

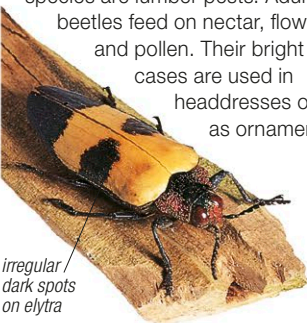
Family Buprestidae

Jewel beetles

Occurrence 15,000 spp. worldwide, primarily in tropical regions; in sunny glades, on bark or flowers



Most jewel beetles are a brilliant, metallic green, red, or blue, with spots, stripes, and bands. They are 1/16–2 1/2 in (0.2–6.5 cm) in length and taper toward the rear. They lay eggs in wood—some species have heat sensors that can detect freshly burned areas where they may mate and lay eggs—and the larvae are typically wood feeders. Many species are lumber pests. Adult beetles feed on nectar, flowers, and pollen. Their bright wing cases are used in headaddresses or as ornaments.



CHRYSOCHROA BUQUETI

This beetle is found in tropical forests of Vietnam and Thailand and is highly prized by collectors. In places, this has led to a decline in local populations.

Family Cerambycidae

Longhorn beetles

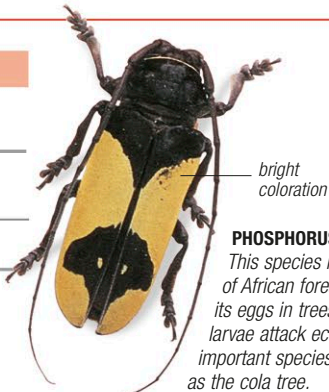
Occurrence 30,000 spp. worldwide; on flowers, at sap flows



Also known as timber beetles, these insects are serious pests of wood. Their larvae, responsible for the damage, tunnel through living trees, and also through wood in buildings. Once adult, they feed on sap, pollen, nectar, or leaves. Adult beetles have strong, toothed mandibles, parallel-sided bodies, and are 1/8–6 in (0.3–15 cm) long. They are often brightly colored, but their most striking feature is their antennae, which may be up to 4 times as long as the body. Females lay their eggs singly under bark. Like many wood-boring species, the life cycle can take several years to complete.

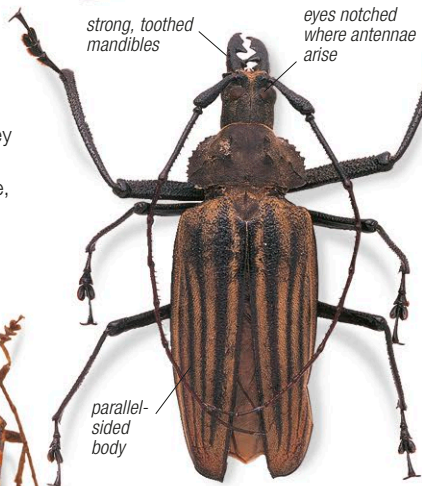
STERNOTOMIS BOHEMANNI

This Afrotropical species has the typical longhorn shape. The females use their mandibles to cut through bark, under which they lay eggs.



PHOSPHORUS JANSONI

This species is a native of African forests. It lays its eggs in trees, and its larvae attack economically important species such as the cola tree.



XIXUTHRUS HEROS

This large longhorn beetle is 3 1/4–4 in (8–10 cm) long and is endemic to Fiji. The males are territorial.

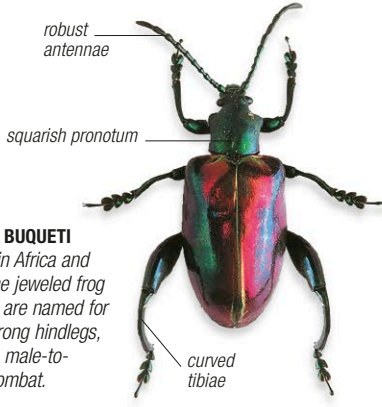
Family Chrysomelidae

Leaf beetles

Occurrence 35,000 spp. worldwide; widespread on most types of plants



Although some eat pollen, adults in this family are mostly leaf eaters—hence the name. This makes many species serious pests, although a few have been used in biological control of weeds. Leaf beetles range from cylindrical and elongated, to round-backed. They measure 3/16–1 1/4 in (1–3 cm) in length, with the antennae less than half the body length. Many species are brightly colored. Eggs are laid in groups on host plants or in soil.



SAGRA BUQUETI

Found in Africa and Asia, the jeweled frog beetles are named for their strong hindlegs, used in male-to-male combat.